

Peeking through the window: Fingerprinting Browser Extensions through Page-Visible Execution Traces and Interactions

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Content

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Peeking through the window: Fingerprinting Browser Extensions through Page-Visible Execution Traces and Interactions Shubham Agarwal Aurore Fass Ben Stock CISPA Helmholtz Center for Information Security Information Security Saarbrücken, Germany Saarbrücken, Germany Saarbrücken, Germany shubham.agarwal@cispa.de fass@cispa.de stock@cispa.de

ABSTRACT Traces and Interactions. In Proceedings of the 2024 ACM SIGSAC Conference Browser extensions are third-party add-ons that provide myriads on Computer and Communications Security (CCS '24), October 14–18, 2024, Salt Lake City, UT, USA. ACM, New York, NY, USA, 15 pages. <https://doi.org/10.1145/3658644.3670339> of features to their users while browsing on the Web. Extensions often interact with the websites a user visits and perform various operations such as DOM-based manipulation, script injections, and so on. However, this also enables nefarious websites to track their visitors by fingerprinting extensions. Researchers in the past have shown that extensions are susceptible to fingerprinting based on the resources they include, the styles they deploy, or the DOM-competitions they perform. Fortunately, the current extension ecosystem contains safeguards against many such known fingerprinting techniques such as Canvas APIs [4], WebGL APIs [5], and other side channels. We present the first study to investigate the fingerprinting characteristics of extension-injected code in pages' JavaScript namespaces to effectively harvest client-side

information specific to individual page and through other observable side-effects like changed cookies. users on the Web and further track their activity across websites. Doing so, we find that many extensions inject JavaScript that pollutes the applications' global namespace by registering variables. It fingerprints vectors for these trackers in recent times, owing to which also enables the attacker application to monitor the execution of their unique position in the overall Web ecosystem. Extensions can inject code by overwriting the JavaScript APIs and capturing perform privileged operations implemented in their background execution traces through the stacktrace, the set of APIs invoked, script or even execute code in the Web applications' execution etc. Further, extensions also store data on the client side and per-context through the content scripts. Tailored to provide a specific form event-driven functionalities that aid in attribution. Through set of features to their respective users, browser extensions may also, in our tests, we find 2,747 Chrome and 572 Firefox extensions to be inadvertently reveal personal information about their users, such as susceptible to fingerprinting. Unfortunately, none of the existing as their geolocation, background, ethnicity, or social and personal defense mechanisms prevent extensions from being fingerprinted interests [24]. These extensions perform a specific and often highly through our proposed vectors. Therefore, we also suggest potential privileged set of operations, such as DOM-based modifications, as a measure for developers and browser vendors to safeguard the changes to cookies, or script injections, to implement their intended extension ecosystem against such fingerprinting attempts. functionality at runtime. These operations, however, could also expose them to being uniquely identified by online trackers. CCS CONCEPTS Prior work has shown different ways of uniquely identifying • Security and privacy Web application security. browser extensions. Some of these include Web-accessible resources-based (WAR-based) fingerprints [47], the side effect of code bloat-keywords [52], behavior-based fingerprints [24, 49, 53], user-induced side Client-side Security, Extension Fingerprinting, Browser Extensions effects [48], and stylesheet-based fingerprints [28]. On the other ACM Reference Format: hand, several mitigation techniques also intend to thwart any fingerprinting attempts based on the above techniques [25, 46, 59]. The window: Fingerprinting Browser Extensions through Page-Visible Execution studies have led to a cat-and-mouse game of newly introduced fingerprinting vectors and subsequent defense mechanisms. A recent Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation for extensions [39] to thwart WAR-based fingerprinting. on the first page. Copyrights for components of this work owned by others than the ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission from ACM. ACM ISBN 979-8-4007-0636-3/24/10 done carefully, this can leave traces of the extension's execution in <https://doi.org/10.1145/3658644.3670339> the

JavaScript namespace of the document, i.e., within the reach CCS '24, October 14–18, 2024, Salt Lake City, UT, USA Shubham Agarwal, Aurore Fass, & Ben Stock

of a fingerprinting attacker. These range from client-side storage (e.g., `localStorage` or cookies) through the invocation of global APIs and properties (e.g., `Array.forEach`), and setting global variables to observable events caused by extension-sent `postMessages`. Notably, none of the existing defenses (i.e., Parallel DOM [25], Shadow DOM [28], or randomized extension URLs [46]) protect against these vectors due to shared window context. In this study, we first discuss how an extension is susceptible to fingerprinting based on its execution trace and other JavaScript-observable side-effects (e.g., changed cookies or sent `postMessages`). We then build a fingerprinting page that sets up the JavaScript environment such that it can capture an extension's interaction with it. Doing so repeatedly allows us to detect which observed functionalities are consistently present to deterministically infer the presence of an extension. We run our analysis on a set of up- Figure 1: The access-control, capabilities, and isolation to-date Chrome extensions, showing that 2,747 extensions can be boundaries of different components in browser extensions fingerprinted through our identified vectors, affecting over 169M users who installed these fingerprintable extensions. Over 59% of the reported extensions adhere to the ManifestV3 standards, highlighting the fact that the issues are a threat to modern extensions. browser vendors, through which they can perform various operations. Moreover, our results transfer to the Firefox ecosystem, where we find 572 extensions that can be detected. Notably, by comparing bookmark management, tab customization, text assistance, password management, or ad-blocking functionalities to their users to detect 1,355 extensions, but importantly would still be able to across different browsing platforms. detect 484 extensions their approaches would be unable to detect if Figure 1 shows different extension components and their isolated dynamic runtime URLs were deployed. Our findings highlight that extension boundaries. An extension includes a mandatory `manifest.json` file, describing an extension's metadata, e.g., the API & host permissions but also add significant fingerprinting surface, which existing missions it holds, as well as the scripts and other Web resources (and proposed) approaches could not readily defend against. required by an extension for seamless execution. The privileged We summarize the key contributions of this study here: component, i.e., the service worker (background script, in the pre- • We identify and leverage two classes of fingerprinting vectors: execution traces and JavaScript-observable side effects scripting or cookies, allowing extensions to inject scripts or get that an attacker can abuse to detect browser extensions in-access to the cookies of the visited page, respectively. It runs in an isolated JavaScript namespace, can communicate through messages, • We then build a dynamic analysis pipeline, Raider, to analyze and has the unidirectional ability to inject scripts into a page. The all free extensions in the Chrome Web Store and identify content script is less privileged since the JavaScript executes in the those that are fingerprintable through our proposed vectors. context of the visited Web page, although in a separate JavaScript We show that 2,747 extensions are uniquely identifiable. namespace from that of the page. It gets a clean reference to the • By applying our techniques to the Carnus dataset, we show- page DOM and can perform read and write operations through case that our techniques can overcome randomized WAR the APIs available in its own JavaScript namespace [34]. This is URLs. Further, our findings for Firefox

highlight that the implemented to ensure that a malicious page cannot abuse the underlying issues exist in both major extension ecosystems. higher level of privileges of the content or even background scripts. • To facilitate future research, we will open source our analysis Notably, while the actual APIs are in a separate namespace, the con- pipeline and the associated dataset [45]. tent script shares client-side stores such as cookies or localStorage with the page. Additionally, extensions can inject scripts into the 2 TECHNICAL BACKGROUND page itself, either by programmatically creating and adding them to the DOM from the content script, or by calling executeScript In this section, we provide an overview on the extensions' archi- from the background [14]. Importantly, the injected JavaScript then tecture and explain the key concepts relevant for our study: the executes with the same privileges and in the same context and (shared) global namespace in JavaScript, available client-side stor- JavaScript namespace as the Web page. age mechanisms such as cookies, and event-driven communication An extension requires corresponding permissions to carry out through postMessages. privileged operations in its background. For example, it requires the bookmarks permission to create, search, update, or remove book- 2.1 Browser Extensions marks from the browser of their users [11]. This, however, is not Browser extensions are client-side add-ons, typically designed by the case with content scripts that have access to all the Web APIs third-party developers to provide additional features to Web users. also available to a Web page by default. For instance, an extension Extensions have access to the powerful Chrome APIs, exposed by can invoke any IndexedDB APIs without requiring any permissions. Peeking through the window: Fingerprinting Browser Extensions through Page-Visible Execution Traces and Interactions CCS '24, October 14–18, 2024, Salt Lake City, UT, USA

```
1 window.foo = "bar"; extensions can also set or get cookies in their background through 2 //
3 Object.defineProperty(window, 'foo', {
4   get: function() { return "baz"; },
5   configurable: false
6 });
7 console.log(window.foo);
```

Overriding the default accessor of the defined variable. 3 Object.defineProperty(window, 'foo', { the cookies permission and its corresponding APIs. 4 get: function() { return "baz"; }, Web Storage API: The Web Storage API was introduced with 5 configurable: false the HTML5 standards and enabled applications to store compara- 6 }); 7 console.log(window.foo); tively large chunks of structured data, as {key: value} pairs, on the 8 //Expected output: "bar", Actual output: "baz" client side [32]. This API enforces origin-level isolation on the data storage and access. One can store data in two different contexts: 1) Listing 1: The dynamic nature of JavaScript allows to cus- temporary data stored only for the current page session, using the tomize the default behavior of built-in APIs & properties. sessionStorage API, and 2) data stored to persist across sessions, using the localStorage API. While sessionStorage only allows up to 5 MB of data storage per origin, localStorage allows com- paratively more data storage. Websites utilize Web Storage APIs to Depending upon the nature of the API permission, both these com- store data such as user state, runtime configuration, personalization ponents may still require appropriate host permission and restrict settings, or code-caching [54]. Unlike cookies, the data stored with their capabilities to these hosts [15]. The permissions (or host_- these APIs remains on the client, and the browser never implicitly permissions in the ManifestV3 standards in Chrome) key contains sends them to an application server. Extensions operating on a given information on hosts an extension's background can operate on. Web origin can also access the Web Storage APIs, and store data The content_scripts key in the manifest contains script paths keyed to this origin through their content or injected scripts [33]. with corresponding host permissions. The web_accessible_re- IndexedDB API: IndexedDB is a JavaScript-based object-oriented sources key includes the definition of other auxiliary resources database that allows client-side components to store extensive struc- (e.g., CSS)

required for an extension's functionality. tured data that persists across sessions [31]. This API also enforces origin-level access control on the data storage and accesses, sim-

2.2 The Global Namespace in JavaScript

ilar to the Web Storage API. This API allows operations based on individual transactions and executes asynchronously, i.e., without Whenever the browser renders a page, all scripts that operate on blocking other executable code in the event loop. Web applica-

the page share the same global object called window. This means tions may store data using IndexedDB for various purposes, such that both first- and third-party scripts can read each others' global as caching of code, network responses, or other static resources, variables, access global functions defined by each of them, and so shared with the Service Workers API [18]. on. This also allows modifications to the execution environment by any script since JavaScript's built-in functions (e.g., the document used to interact with the DOM or the Array constructor) are also

2.4 postMessages & Other Runtime Events

merely global (automatically initialized) variables. This dynamic Extension components can communicate with each other (i.e., con- nature of JavaScript allows Web developers to overwrite the native tent scripts, web-accessible resources) or with a Web page through definition of nearly all the built-in APIs and properties it offers the postMessage API. Here, the sender, either an extension script or to the Web page. For example, as shown in Listing 1 (lines 3–6), a Web page, sends the message data by also optionally specifying a developer can override the native accessors of the properties the message target [17]. The other party listens to the message defined on the window object to alter its behavior (line 8). The by registering a corresponding message handler. It is pertinent to overwritten behavior of APIs affects all the JavaScript that executes note that since any extension-initiated postMessages execute in the in the same global namespace. Notably, while the example shows context of the application, the effective origin of these messages is how to overwrite a getter of a specific property, a developer can the Web page's origin. Thus, the Web page can also register a cor- easily overwrite existing functions as well. Recall that the content responding event listener and listen to all the message exchanges. script runs in an isolated namespace, whereas the scripts injected Similarly, a content script may also dispatch custom events to a page. into the page by the extension (either from content or background While it is infeasible to a priori know which events may be fired, scripts) share the page's namespace. an extension's injected script still needs to register a listener for said events, which can be observed due to the necessary invocation

2.3 Client-Side Storage

Mechanisms of the addEventListener function in the global scope. We now briefly describe each client-side storage API accessible by both the Web applications and the extensions.

3 THREAT MODEL

Cookies: Cookies used to be the traditional way of storing data We consider an attacker who is capable of having a victim visit on the client side, associated with respective hosts, before the lo- their website but who cannot control a specific website. That is calStorage and sessionStorage APIs were introduced. However, if an extension only operates on Facebook, this is outside of our modern Web applications still use cookies to store information on threat model, as the adversary cannot gain control over that site. users' machines. Specifically, a browser sends cookies with all the More concretely, any traditional Web attacker may try to detect the outgoing requests to a particular host. Extensions can set, delete, or existence of one or more target extensions installed on the client modify cookies on a visited Web page through scripts executing in side. The attacker can use the information associated with these the context of a Web application (i.e., content scripts and WARs) us- extensions to infer privacy-sensitive characteristics of their visitors, ing document.cookie, similar to native applications. Additionally, such as their geo-

location, ethnicity, or religion [25]. The attacker CCS '24, October 14–18, 2024, Salt Lake City, UT, USA Shubham Agarwal, Aurore Fass, & Ben Stock

can further share this information with third-party trackers or even 1 // content_scripts.js 2
localStorage.setItem('foo', 'bar'); use it themselves, all without the users' knowledge or consent. 3 //
attacker-webpage.js Browser extensions can interact with and store data on the client 4 for (let
index = 0; index < localStorage.length; index++) { side using storage APIs (e.g., localStorage) via
content scripts 5 6 let key = localStorage.key(index); let value = localStorage.getItem(key); and
other web-accessible resources, as also shown in Figure 2a 7 logStorage(key, value); (line 2). In this
case, an attacker application can detect any target 8 } extension(s) installed on the client side by
probing for items within (a) Storage APIs scanning these data stores, accessible through the
application JavaScript (Figure 2a, lines 4–8). Unfortunately, none of the existing defenses 1 against
fingerprinting prevent these observable side effects. 2 // content_script.js
window.postMessage('Hello from CS!', '*'); Next, as shown in Figure 2b, extension components may
com- 3 // popup.js municate with Web pages or even among themselves (i.e., popups 4
window.addEventListener('message', function (event) { 5 event.source.postMessage('Message
received!'); or WARs) via postMessages (line 2). They could also register other 6 }); event listeners to
execute event-driven operations (e.g., scrollup, 7 // attacker-webpage.js onmouseover, other
custom events) either through content scripts 8 window.addEventListener('message', function
(event) { 9 logMessages(event.data); or the injected code (lines 3–6). Here (as shown in lines 7–10),
the 10 }); attacker JavaScript can also listen to all the postMessage exchanges issued by the
extension scripts since these scripts are injected and (b) Intercepting postMessages executed in
the same context. Similarly, the page JavaScript can forcefully intercept or trigger runtime events
registered by the ex- 1 // injected-script.js tensions to induce observable side-effects, as they share
the same 2 extension_key = "extension_value"; 3 // attacker-webpage.js execution context. In this
case, the attacker must know the target 4 for (let prop of Object.getOwnPropertyNames(window))
{ events a priori to trigger them and cause any observable side effects. 5 logProperty(prop,
window[prop]); 6 } As discussed in Section 2, extensions can inject JavaScript into the visited page,
either from the content script or through the back- (c) Global variables set by extensions ground.
In this case, the injected JavaScript executes in the same context as the Web page, similar to the
content script. However, in Figure 2: Observable Behavior of Extensions at Runtime addition, the
injected code also shares the global JavaScript names- pace with the Web page, such that its
execution may cause side effects to this namespace. For instance, as in Figure 2c (line 2), the 4
RESEARCH METHODOLOGY injected script may register event listeners, set variables in the global
scope, and access/modify native functions and properties Our overarching research question is:
how many extensions can be directly accessible to the Web page JavaScript. In this case, the
uniquely fingerprinted through the traces they leave in the global attacker can monitor the usage
of these global APIs to detect the scope of a visited page or through their otherwise visible side
effects? execution traces of extension-injected code. For example, as in lines To answer this, we
first need to identify extensions that include 4–6, the attacker JavaScript can iterate over all global
properties scripts or have permissions that may allow them to store client-side on the window to
detect extension-defined ones. Thus, extensions data, send postMessages, or inject scripts into
the page. We do so by that inject JavaScript into arbitrary Web pages are susceptible to statically
analyzing the manifest files and filtering out extensions fingerprinting through the execution
traces of the injected code. without the necessary permissions. Subsequently, for the remaining

We assume our attacker to be sufficiently able to download all extensions, we need to learn which of them use the capabilities browser extensions from the extension store [10] and run offline at runtime. For this, we spawn browsers to load each extension analysis on them to observe their behavior and derive identifiable and visit our test page, allowing us to capture the extension's inter- signatures. The attacker could then use these signatures to detect actions with it. To then answer our main question, we determine installed extensions and, subsequently, their users online. Notably, traces (observable side-effects) unique to each extension. the attacker only needs to run the offline analysis step for new extensions on the store or when an existing one is updated. Here, 4.1 Raider: Overview we only consider extensions that operate on any Web pages for our We build an automated dynamic analysis framework, Raider, to analysis, such that an arbitrary attacker application can fingerprint answer the above question and detect extensions based on a.) the them. However, it is pertinent to note that extensions that run only execution traces on the global JavaScript namespace from extension- on specific pages may still be identifiable by the corresponding injected scripts; and b.) their side effects through interactions with applications through any of the vectors proposed in our study client-side Storage APIs or sent postMessages. Figure 3 depicts the (meaning that the results presented in this paper are a lower bound high-level overview of our methodology. We start by i.) unzipping of the extensions we can fingerprint with our approach). the extensions and statically analyzing their manifest to identify scripts that operate on <all_urls>. ii.) We then extract all valid content scripts and web-accessible resources as JavaScript that these extensions declare in their manifest and determine host permissions for individual scripts. iii.) We also check for extensions' capabilities Peeking through the window: Fingerprinting Browser Extensions through Page-Visible Execution Traces and Interactions CCS '24, October 14–18, 2024, Salt Lake City, UT, USA

Figure 3: Overview of methodology: i) Extract package and parse manifest. ii) Check for valid content scripts and WARs that operate on all URLs. iii) Check for script-injection capabilities from the background, iv) Select those extensions that have script-injection permissions from either (ii) or (iii). v) Spawn browser instance and load extension. vi) Navigate to the test pages. vii) Collect signatures and store them. viii) Analyze the uniqueness of signatures to confirm fingerprintability. to inject JavaScript from the extension background. iv.) Lastly, we or send postMessages, and thus, require further scrutiny to de- select those extensions for our next stage where an extension either termine if they are fingerprintable, based on our threat model. a.) has at least one content script or WAR operating on any URL More specifically, our tool parses the manifest of individual ex- (i.e., <all_urls>); or b.) has permissions to inject JavaScript into tensions to detect any content_scripts or web_accessible_re- arbitrary hosts from its background. sources declarations to extract relevant JavaScript files. This is With the selected set of extensions, the pipeline then analyzes because an extension can only interact with the storage APIs or their runtime behavior to collect fingerprinting signatures. The send postMessage from these scripts. Moreover, content scripts also dynamic step (v.) – viii.) involves two different data collection enable the injection of other scripts into the page. Extensions may strategies: we collect the execution traces of the extension-injected also choose to execute JavaScript in the page's context from their code on the global JavaScript namespace differently from the way background/service worker at runtime without declaring them in we capture the extension-driven interactions with the Storage and their manifest, using the scripting or tabs API [14]. This per- postMessage APIs. We do this by loading extensions individually and mission also allows extensions to inject or update content scripts

capturing any side effects they cause on the global namespace as from their background at runtime. In the Manifest V2 standards, this signatures through our specially crafted test page. Then, we collect translates to the tabs permission [14]. Additionally, extensions may all the data set by individual extensions in any of the client-side also set cookies on arbitrary domains from their background/core data stores at runtime by individually loading them in the browser through the chrome.cookies API or even inject the Set-Cookie instance and polling these data stores periodically through another header within HTTP response headers, by requesting the webRe- test page. After collecting all data, we determine whether these quest or declarativeNetRequest permissions. While extensions extension-driven interactions are distinct and appear consistently. can interact with the client-side storage, register event listeners, or To that end, we visit the test page nine times and only consider cause side effects to the global namespace through different scripts, behavior to be relevant if it is observed in all nine visits and is they should also operate on all URLs for any attacker application unique to one extension. to fingerprint them. Thus, we only consider those extensions with sufficient host permissions that allow them to run on arbitrary hosts, i.e., <all_urls> and equivalent. We extract the host permis- 4.2 Static Pre-filtering sions specified for the content scripts, web-accessible resources, The first stage of our proposed methodology consists of a static and background scripts, remove any wildcards, and normalize them extension pre-filtering step. Here, we identify those extensions that to detect their actual operational set of hosts. This is necessary as can inject JavaScript into the DOM, interact with storage APIs, developers may specify match patterns instead of fully-qualified CCS '24, October 14–18, 2024, Salt Lake City, UT, USA Shubham Agarwal, Aurore Fass, & Ben Stock

domain names [8], (e.g. ://, or http://foo*) which may still 1 function _hook(object, property, api) { 2 // Preserving native definition of the function. effectively allow extensions to operate on arbitrary Web pages. 3 let __originalFunc = object[property]; Note that we intentionally skip extensions that only have the ac- 4 // Custom definition for Global APIs tiveTab permission or optional_host_permissions (for MV3) 5 function __customFunc() { 6 // Extracting API related information. in their manifest. While this gives an extension the same capabilities 7 let context = this; as an explicit host permission, it requires the user to actively en- 8 let args = Array(...arguments); gage with the extension. This is outside of our threat model, which 9 // Extracting the source code of the executing code. 10 let callerData = {}; does not require user intervention outside of the page. Appendix A 11 let caller = arguments?.callee?.caller; shows an example of extensions with valid script declarations in 12 while (caller) { 13 callerName = caller.name; the manifest relevant to our study. As shown, content_script.js will 14 callerFunc = caller.toString(); execute on all HTTPS URLs, while storage.js and cookies.js operate 15 callerData[callerName] = callerFunc; on all URLs, irrespective of the URL scheme. Thus, any website 16 caller = caller?.arguments?.callee?.caller; 17 } running on HTTPS can be a potential adversary in this case, as per 18 // Capturing the stack trace of the executing code. our threat model described in Section 3. We note that applications 19 let stacktrace = new Error().stack; may also directly send messages to the extension core using the 20 // Sending data to our test server. 21 logToServer({ api, context, args, stacktrace, callerData }); chrome.runtime API [13]. However, this is only possible when the 22 // Now, returning the result from executing native function. extension intentionally opts-in to be reachable from a given website 23 return __originalFunc.apply(this, arguments); 24 } by specifying the respective domains as externally_connectable 25 // Replacing the native definition with custom definition. in their manifest [7]. Hence, we discard such extensions here. 26 object[property] = __customFunc; To sum up, we select extensions for our analysis, that: i.) have 27 } 28 at least one valid content script or web-accessible resource running 29 //Instrumenting APIs now... on

<all_urls>, or ii.) have valid background script running on <all- 30 __hook(Array.prototype, "forEach", "Array.forEach"); urls>, and request for any of the following permissions: scripting, tabs, cookies, webRequest, or declarativeNetRequest. Listing 2: Logic to overwrite globally-accessible APIs.

4.3 Execution Traces of Injected Code 4.3.1 Overwriting global APIs. We begin by discussing the steps An extension's content script can interact with the given page to overwrite the JavaScript APIs and the global property accessors, through the window and the document handle. These two objects, enabling us to capture their invocations or accesses, respectively. however, are not shared with the page itself. While any changes At the same time, we also intend to preserve the original behavior through document apply to the page's DOM, changes to the con- of the overwritten components to avoid any side effects of our tent script's window object are opaque to the visited page. This way, instrumentation at runtime. For our purposes, this is particularly the content script of an extension cannot, intentionally or acci- important to ensure an extension can fully execute all of its func- dentally, alter the behavior of the JavaScript APIs, properties, and tionality, which provides ample chance to fingerprint it. variables declared by the Web page JavaScript, and vice versa. In or- We demonstrate the steps to perform API overwrites through der to run extension-specified code within the context of the page's Listing 2. Here, the **hook method instruments the individual window object, this code must be explicitly injected into the page. APIs to enable the logging mechanism. Concretely, it first stores Extensions can perform script injections either directly through the original definition of the API under instrumentation (as __- their content script – by using the document.appendChild API, or originalFunc in line 3). Then, it defines a custom method (here, through their background script – by using the scripting (or the __customFunc as in lines 5–24) where we define the logic to capture tabs) permission (chrome.scripting.executeScript). all the relevant invocation-associated details. It also includes the Now, the extension-injected code could perform a wide array logic to dispatch the collected data to our test server (line 21). Once of operations in the context and the namespace of the visited ap- the logging is complete, this __customFunc returns the result from plication. More importantly, the JavaScript APIs the injected code the native definition of the called API through __originalFunc may utilize or the window properties that the injected code may (line 23). Thus, our custom definition does not affect the natural read or write during its execution are also shared and observable execution flow of the injected code. In the end, we overwrite the to the Web page through its JavaScript. This behavior enables an native definition of the API with our custom-defined logic (line 26). attacker Web application to observe, or even further, to overwrite This way, we instrument a total of 571 different JavaScript APIs, the native behavior of nearly all the JavaScript APIs and properties, accessible to both the Web page and the extension-injected code in that particular JavaScript namespace to monitor their usage. For in the global JavaScript namespace (similar to line 30) [38]. We example, an attacker JavaScript can overwrite the native definition follow similar steps to overwrite the native definition of JavaScript of the Array.prototype.forEach API to actively observe any in- property accessors, such as document.title and window.name. vocations of this API. If an extension-injected code now invokes the Here, we use the __lookupGetter and lookupSetter APIs forEach operation on any array, the attacker will then be able to to obtain the native definition of the property accessors. We then observe its invocation. This capability of the Web page's JavaScript overwrite them using the Object.defineProperty API. This way, code can be leveraged to detect an extension's behavior in**

various we overwrite 51 other globally accessible JavaScript properties. We ways, which we discuss in the following. selected the global JavaScript APIs and properties that are standard Peeking through the window: Fingerprinting Browser Extensions through Page-Visible Execution Traces and Interactions CCS '24, October 14–18, 2024, Salt Lake City, UT, USA

built-in JavaScript objects [38].¹ Now, we elaborate on the relevant 1 //Client-side storage state as polled on every 0.5 second for 10 invocation-associated details from these instrumented APIs and times after page load. 2 window.addEventListener('load', async function (e) { the properties we extract. 3 let counter = 0; 4 setTimeout(async function run() { 4.3.2 Relevant contextual data from API invocations. Our instru- 5 if (counter++ < 10) { mentation allows us to capture different sets of data based on the 6 await pollStorage(); 7 setTimeout(run, 500); nature of the invoked API. For instance, as shown in the first ex- 8 } ample of Appendix A, when the Array.forEach API is invoked on 9 }, 500); foo, the value of this in the current execution context is the array 10 }); 11 // Polling data stores before page is unloaded/navigated away. passed to the API (i.e., [1, 2, 3]). The argument is the callback that 12 window.addEventListener('beforeunload', async function(){ processes the iterated element. Here, suppose an extension-injected 13 await pollStorage(); code invokes this API. In that case, our logger will capture three 14 })

critical pieces of information: the API name, the arguments passed, and the value of this in the execution context (Listing 2 – lines 7 Listing 3: Polling different data stores on page events and on and 8). There are other APIs that are static methods of their parent specified intervals to log data. class and do not have their own context or this value (e.g., Array.isArray in Appendix A). Here, we only capture the name of the invoked API and its arguments. filename, line number, and offset. As we discuss in Section 5, both are distinctly unique features across a vast amount of extensions.

4.3.3 Obtaining the source code of the injected code. Whenever an executing script invokes a function, fn, the invoked function 4.3.5 Capturing global variables. To avoid polluting the global fn also contains the pointer to its caller [36]. That is, the invoked namespaces, functionality can be wrapped in an immediately-invoked function has the pointer to the invoking function through the argu- function expression (IIFE). However, if the code injected by an exten- ments.callee.caller property. This is also true for the JavaScript sion either (a) does not use an IIFE, (b) defines a variable without a APIs and the property accessors in the global namespace that we var keyword, or (c) explicitly sets window.foo, this results in a glob- consider in this study. Through this, an attacker Web page could ally accessible variable. To detect these variables, the attacker Web extract unique caller functions or even leak the entire source code page can enumerate all the properties available on the global scope. of the script and use them as vectors to detect extensions later. We collect all the identifiers (i.e., variables and function identifiers) Concretely, suppose an extension-injected code invokes any in- that extension-injected code writes on the JavaScript namespace of strumented APIs or accesses any property we instrument. In that our test page. Here, we utilize the Object.getOwnPropertyNames case, our custom method also recursively extracts the caller of the API to enumerate all the properties on the window handle and dis-invoked API until it is set to null. Lines 9–17 in Listing 2 show card those that are artifacts of our test page or are also seen in our approach to collect caller-associated details for an invoked API. the extension-less environment (e.g., browser built-in APIs). This Notably, if the caller is a.) top-level code; b.) an arrow, async, or way, an attacker can probe for variables associated with individual generator function; or, c.) runs in the strict mode, it is always set to extensions to check for their presence on the client side. null, and our logger cannot capture anything. Note that

reading the source code of injected scripts could also be done through a 4.4 Side Effects: Storage APIs and Messages MutationObserver; this, however, can be easily defeated through a Besides direct changes to the global JavaScript scope and variables, ShadowDOM [28], which is why we do not consider this vector. extensions can cause other side effects which can be polled for or 4.3.4 Capturing the stacktrace. The arguments.callee.caller listened to from JavaScript. does not always return the handle to its caller, primarily in cases 4.4.1 Cookies, LocalStorage, and IndexedDB. In line with the secu- where the entire injected code is running in the strict mode, or the rity model of extensions, the content scripts do not share the same API invocation occurs on the top-level code. However, the attacker namespace with the Web page when accessing storage and cookies. can still capture the execution stack of the injected code up to the That is, an extension can invoke document.cookie from the con- point where the API invocation occurs. This stacktrace not only tent script to set a cookie for the page, yet the invocation is not di- provides the names of the functions called in reverse order but rectly observable by the JavaScript running on the Web page. How- also contains both the URL of the file (if the code is in an external ever, the underlying storage/cookie values are shared, i.e., the effect script) and the line number and offsets. We collect this information of a newly added cookie can be observed from the page's JavaScript by accessing the stack property of the Error object (as shown realm. The same applies to both localStorage and sessionStor- in line 19 of Listing 2). In cases of randomly generated runtime age as well as IndexedDB. Note that in Firefox, IndexedDB cannot identifiers (both in Chrome and Firefox), the filename for scripts be polled without prior knowledge of the name of the database since included as web-accessible resources contains a random identifier. there is no implementation of the indexeddb.databases API [37], Therefore, we consider two attacks: the full stacktrace (including which is why we do not collect any data for it. To observe values runtime identifiers) and a normalized stacktrace (for a hypothetical injected by the extension, the attacker simply polls the storages to case of widely adopted randomized runtime identifiers) for which see whether they contain any content. This is shown in Listing 3, we remove the extension IDs from the trace; leaving us with the where the attacker's code iterates over all storages every 500 ms to 1 Please visit our repository for the complete list of hooked APIs and properties [45]. see if the extension stored any data. For our purposes, we do not CCS '24, October 14–18, 2024, Salt Lake City, UT, USA Shubham Agarwal, Aurore Fass, & Ben Stock

1 non_unique_features = set() leave any trace that an attacker might observe. We perform this run
2 3 unique_exts = set() for feature, extension_ids in feat_to_exts.items(): for a total of three times
for each of these extensions that exhibited 4 if len(extension_ids) == 1: such a behavior. This way, if
a feature occurs repeatedly, it cannot 5 unique_exts.update(extension_ids) be due to random
chance but is instead deterministic. 6 else: 7 non_unique_features.add(feature) We first extract all
the API invocations, variables, messages, etc., 8 for each extension in the dataset and aggregate
the number of oc- 9 for id1, id2 in itertools.permutations(non_unique_features, 2): currences (here
dubbed visits) in which the feature was observed. 10 intersection = feat_to_exts[id1] &
feat_to_exts[id2] 11 if len(intersection) == 1: Subsequently, we iterate over all the features to
identify those that 12 unique_exts.update(intersection) occur repeatedly. For each such feature,
we use it as the dictionary key to store those extensions that use the given feature (repeatedly).
Listing 4: Detecting uniquely identifying features From this dictionary, we can already trivially find
all those exten- sions that are uniquely identified by a feature. If, for a given feature, the length of
the set is exactly 1, this feature uniquely identifies look into the exact values being written to the

respective stores but an extension (as shown in line 4 of Listing 4). For all extensions instead focus only on their overall uniqueness. that cannot be detected by a single feature, we try combinations of two features that might be unique to a single extension. Here, 4.4.2 PostMessages. Last, but not least, the content script can di- we iterate over all combinations of features (within each class, so, rectly communicate with the page through postMessage. We note e.g., all cookie names are combined) to see if the intersection of that this vector was already discussed by Karami et al. [24], yet falls the extensions that exhibit that feature is 1. In that case, the at- into the category of JavaScript-visible side effects, which is why tacker who monitors an extension's behavior and observes these we consider it also in our work. Notably, the content script and the two features can conclusively say that the given extension must be page itself receive any incoming postMessage to the page. While, installed. Note that the approach could be expanded to also contain again, the page's JavaScript cannot hook into the addEventLis- 3-tuples of features. However, this significantly increases runtime tener API used by the content script, it can nevertheless register (which is the cubic relative to the number of features), which is why its own event handler to capture all incoming messages. This way, we did not consider this in our work. Moreover, experimentally, if an extension's content script sends a message to the other com- we could verify that all but two extensions in our dataset were ponents, this can be recorded by the attacker's script. In our initial fingerprintable through only a single feature within the same class. experiments, we found that while the exact message content often varies (e.g., because of timestamps or randomized values), the keys 5 EVALUATION & RESULTS of messages (when using JSON messages) remained stable. With our framework, we now perform an analysis of three different datasets to showcase the potency of our attacks. For this, we first 4.5 Data Collection and Identification collected all the free extensions from the Chrome Web Store and To test an extension, we install its crx file in a fresh browser instance Mozilla Add-ons Store, available as of January 3rd , 2024. We refer and visit our specially crafted test pages. Here, the first test page to them as Raider and Firefox, respectively. Then, we also gathered constitutes hooks, as described in Listing 2, and captures the execu- the dataset from Karami et al. [24], referred to as Carnus, along tion traces of any extension-injected code. In the second test page, with the fingerprinting labels from the original findings. In our we poll individual data stores for any data and enumerate global study, we use these datasets to run our experiments and understand variables set by extensions, as in Figure 2. Since the test pages and the trend of fingerprinting behavior in the extension ecosystem. the tools used by prior works are not publicly available [24, 48, 53], Table 1 shows an overview of the datasets we use. Note that our pre- we could only obtain a prototypical honey page used by Karami filtering step to identify extensions that do not have the necessary et al. [24]. We include all the elements (e.g., iframes, audio/video permissions reduces the total number of extensions to consider tags, etc.) from the Carnus honey page. We further enhance our test further. In particular, the largest datasets are ours (Raider) and the pages by triggering a wide range of mouse and keyboard events on one from Carnus with almost 40k extensions each; Firefox contains page load, corresponding to Table 1. in [48], through dispatching less than 10k extensions. In the following, we analyze those datasets a series of JavaScript events. Here, we dispatch keyboard events separately: first our Chrome dataset Raider, which we subsequently for all possible keys and their hotkey combinations. Similarly, we compare with Carnus. Finally, we consider Firefox. also send mouse events for different elements (i.e., text-selections, For each dataset, we first conduct two runs on the entire dataset. image, form fields, etc.). Naturally,

extensions that do not react to In each run, we visit our test page three times. This total of six page JavaScript-induced events (i.e., check the `isTrusted` property of loads is meant to ensure that as many extensions as possible show the event object) will not be triggered. In the end, we collect and dispatch the execution traces to our backend for processing. Our instrumentation and test pages provide us with the ability Dataset Downloaded After Pre-Filter to observe changes that an attacker could also observe. However, Raider 156,997 37,697 extensions may use random variable names or use timestamps Firefox 26,591 9,488 Carnus 104,484 39,890 for keys and values, i.e., a single run does not suffice to identify persistent features of an extension. To account for that, we visit our Table 1: Extension datasets overview pages three times per run for each extension to see which extensions Peeking through the window: Fingerprinting Browser Extensions through Page-Visible Execution Traces and Interactions CCS '24, October 14–18, 2024, Salt Lake City, UT, USA

Method Usage Repeated Unique Only Installs (i.e., removing extension IDs from the traces) still allows us to detect Global APIs 1,878 1,872 1,769 - 109,584,572 1,569 extensions. Note that out of the 397 extensions that could only

- Stacktrace 1,878 1,871 1,753 397 108,382,340
- Norm. Stacktrace 1,878 1,871 1,569 (237) 103,582,524 be fingerprinted through the full stacktrace, 237 would have also
- Caller & Params 1,878 1,868 813 2 32,740,589 been only fingerprintable through the normalized stacktrace. This

Variables 1,730 1,664 1,301 245 67,048,809 is important given that if Chrome was to widely adopt randomized Cookies 201 198 154 78 4,709,235 Storage 634 623 391 266 8,317,933 runtime identifiers for extensions, the vast majority would still be IndexedDB PostMessages 128 1,069 126 1,028 32 737 17 283 1,580,655 38,519,471 fingerprintable due to the unique nature of filenames, lines, and line offsets in the call stacks. Cross-class 1,634 1,610 1,257 0 48,466,020 We note that while the stacktrace is the most potent attack, all of Total 3,398 3,308 2,747 - 169,093,032 the other vectors, except for the parameters and the caller code, add Table 2: Results for the Raider dataset fingerprinting surface. (The Only column in Table 2 represents the number of extensions fingerprintable exclusively through that indi- vidual vector.) Even the rarely occurring IndexedDB vector allows any behavior. We then retain as our dataset for the next step all us to identify 17 extensions that could not otherwise be detected. those extensions that exhibited any attacker-observable behavior Finally, we consider cross-class fingerprints, i.e., those where single at least once. features are insufficient to identify an extension, yet combining Subsequently, we run three more times for each extension that two features from different classes suffice (e.g., a registered vari- exhibited some behavior before to capture the consistent runtime able together with a specific cookie). The fact that 1,257 extensions characteristics across multiple runs. Thus, we visit our test page a can be fingerprinted through cross-class features highlights that total of nine times per extension. However, in reporting numbers extensions frequently exhibit several types of attacker-visible ac- for fingerprintable extensions, we only rely on those that showed tions. Finally, if we were to disregard the `postMessage` vector (as precisely the same fingerprintable feature in all nine visits across already discussed by Karami et al. [24]), 2,464 extensions would be the three runs, i.e., we provide a lower bound. fingerprintable (as extensions are often unique by multiple vectors), showing the impact of our newly proposed vectors. 5.1 Raider Dataset The discovered fingerprinting attacks have an impact on a large user base (based on the installation counts from

the Chrome Web Extension Detection. Table 2 shows our findings on the Chrome Store). Note that the numbers are lower bounds, as the Store pro- dataset2 . 3,398 extensions make use of some browser functionality vides only inaccurate numbers for popular extensions (e.g., 1,000+). that could be observable by an attacker. Over half of them are related Overall, the extensions that our attacker models can fingerprint to global APIs being invoked by the injected code. Notably, 1,730 are installed by a total of over 169M users (please refer to Ap- of these extensions pollute the global namespace with variables. pendix A for more details). The most prominent examples are the Overall, the usage of IndexedDB is very limited (128 extensions), Malwarebytes Browser Guard and MetaMask, each with over 10M yet we see that all of our analyzed features are in use by extensions. users. Both are fingerprintable through their usage of global APIs Notably, as discussed before, not all invocations, storage ac- by unique stacktraces. In particular, Easy Ad Blocker is another cesses, and messages are deterministic. Consider the example of noteworthy example since adblocker blockers can easily detect the postMessages: here, extensions may send a message that includes a presence of the extension. The reported extensions also span across timestamp in a key. This means that the structure of the message 22 different categories, as listed in Appendix A. changes. In our analysis, we found that 1,028 / 1,069 extensions send The number of extensions fingerprintable at least once in any messages with the same structure repeatedly. However, even in that of the runs is slightly higher (2,760). However, the additional 13 case, if two extensions send the exact same message, an attacker extensions did not show the same behavior in all three runs, which cannot tell those two extensions apart. This is highlighted by the is why we exclude them and provide a lower bound. fact that 737 / 1,028 of extensions that send deterministic messages actually send uniquely identifying messages. Multi-Extension Analysis. So far, we only focused on individual Overall, we find that 3,308 / 3,398 (97%) extensions have some extensions and their fingerprintability based on the vectors we features that occur deterministically. Out of those, we can uniquely proposed. However, users often install multiple extensions that may identify 2,747. We also see that the stacktrace is the single most sig- interfere with each other’s behavior at runtime. In turn, this may nificant contributor to identifying extensions uniquely. 1,753 / 2,747 impact the fingerprintability of these extensions in the presence (64%) of extensions that are fingerprintable within their group are of others or, vice-versa, detections of extensions that do not show detectable through the stacktrace alone. Notably, this is signifi- cantly higher than the caller’s code, which initially seems counter- intuitive. However, our manual analysis showed that extensions N 2 3 4 5 6 7 8 9 10 Avg. frequently leverage libraries. These libraries frequently make use TP (%) 99.7 99.4 99.3 98.8 97.5 96.6 96.4 97.5 97.4 98.0 of JavaScript’s strict mode, which disables the usage of arguments, FN (%) 0.3 0.6 0.7 1.2 2.5 3.4 3.6 2.5 2.6 1.9 0.4 0.4 0.4 0.4 0.4 0.4 0.5 0.4 0.5 0.4 thereby rendering the caller attack infeasible. However, this does FP (%)

not turn off the stacktrace. Considering the normalized stacktrace F1 (%) 99.7 99.5 99.3 99.2 98.5 98.1 97.9 98.5 98.4 98.8

Table 3: Multi-extension results (average over five runs) 2 Note that the Total row is not the sum of the other rows, as extensions often exhibit multiple classes of attacker-observable behavior. CCS ’24, October 14–18, 2024, Salt Lake City, UT, USA Shubham Agarwal, Aurore Fass, & Ben Stock any behavior when run in isolation (e.g., because they require a Method Usage Repeated Unique New No WAR

postMessage to trigger functionality). To test the robustness of our Global APIs 894 889 712 37 207

- Stacktrace 894 889 699 37 -

approach, we now analyze the fingerprintable extensions reported - Norm. Stacktrace 894 889 627 37 186 by Raider in two different multi-extension settings. - Caller & Params 894 885 429 31 120 Variables 1,136 979 638 47 208 Setting 1: Fingerprintable extensions only First, for each Cookies 80 78 31 8 15 extension, we combine it with a set of N-1 other randomly selected Storage 423 419 242 93 149 extensions reported as fingerprintable, where N ranges from 2 to IndexedDB PostMessages 15 497 14 474 11 273 3 14 8 34 10, and load them in the browser to visit the test page and collect Cross-class 775 764 474 18 125 the fingerprints. This is analogous to the tests performed by Karami Total 2,119 1,943 1,355 180 484 et al. [24]. We then collect all potentially identifying features (i.e., all of the vectors we consider) and store them in a database. Since Table 4: Results for the Carnus dataset we know the ground truth of extensions that were installed, we then compare if the behavior observed at runtime can be attributed correctly to the extensions actually loaded during the test. Table 3 shows the results for our multi-extension test with different values to be a side-effect of Selenium crashing for some extensions (for of N. We observe that our proposed vectors are able to accurately unknown reasons), the chance of which is exaggerated due to us detect extensions in 98% cases, even when loaded with multiple testing ten extensions in parallel. We collected the information other extensions with similar runtime behavior. Across all different on the successfully loaded number of extensions for each test by values of N, we have an average of 1.9% false-negative cases where navigating to the extension page and enumerating the loaded set Raider cannot detect an extension when present. On manual inspec- of extensions using Selenium. Of these 2,680 extensions that were tion, we found that some extensions execute blocking JavaScript successfully loaded, our tool accurately detected 99.1% of them in code which then delays the execution of the code injected by other the second multi-extension setting (averaged over five runs). For extensions and our tool fails to capture their invocations after 30 the remaining 0.9%, the side effects from other extensions masked seconds. In other cases, we saw that some extensions also mask the behavior that allowed us to fingerprint them in the single- the behavior of others through their operations (e.g., by themselves extension case. We note that this may relate to our test pages loading hooking into APIs, and thus, hindering attribution through the significantly slower in the presence of multiple extensions. stacktrace). Similar to the previous multi-extension setting, we found 0.5% We also measured an average of 0.4% falsely-labeled cases where false-positive cases where we detected an extension that we did not our tool detected an extension that was not loaded in the tests (note load for the tests. Here, the falsely detected set of extensions and the that we count this relative to the number of fingerprintable exten- false-positive rate are in line with the previous multi-extension set- sions). We investigated this further and found that many extensions ting. Overall, the fingerprinting rates of extensions across different react to the operations executed by other extensions during the multi-extension settings are similar. Since we randomly sampled tests (e.g., code injections, global variables, postMessages, etc.) and almost our complete dataset (i.e., 34,774 / 37,697 extensions with thus, create new but overlapping execution traces for extensions permissions to interact with the page across five runs) for our ex- that are not installed, which eventually leads to false attributions. periments, we do not believe that the false-positive rate would For instance, one extension set the global variable (web3) only in be significantly higher across other combinations of extensions the multi-extension tests but not during individual tests. Since this installed by the

user. variable was only observed by one other extension in the single- extension tests, we falsely flagged said extension as being detected.

5.2 Carnus Dataset Setting 2: All candidate extensions Orthogonally to the previ-

As a second dataset, we rely on Carnus' [24] set of extensions our case, which assumes a user would install N extensions out of a we received from the authors. Fortunately, the authors provided small set of the fingerprintable ones, we also investigated the case us with the complete test dataset, i.e., both fingerprintable and of randomly choosing nine other extensions from the 37k exten- non-fingerprintable extensions, as reported by them. This labeled sions which could potentially interact with the page (see Table 1). dataset allows us to see how many additional extensions our attacks We instantiate fresh browser instances with these 10 extensions can fingerprint on top of Carnus' methods. Overall, Carnus can installed and then perform our tests. We do not select an extension detect 29,428 extensions, the vast majority of which is identifiable more than once to cover as many extensions in our tests as possible. through WAR-based methods (25,866). We note that since their As before, we collect the data and analyze it in the backend to deter- work, browsers have introduced the option to enable dynamic URLs mine the uniqueness of the collected signatures. Note that not all for WARs [39]. This means that extensions can opt to no longer use extensions that fulfill the static requirements in their manifest can a deterministic identifier for the WARs but, instead, a randomized actually be loaded without error. Therefore, in some cases, less than one that resets on reloading the extension or restarting the browser. 10 extensions were loaded. As with the previous multi-extension If this is set, an attacker can no longer probe for specific resources, tests, we perform these tests five times here as well with different as the random runtime identifier is not mapped to an extension. combinations of extensions. In line with our approach for Raider, we confirm that exten- Our tests indicated that out of 2,747 extensions, which were sions are fingerprintable in all three repetitions of the analysis run. consistently fingerprintable (see Table 2), on average 2,680 (98%) We find that 1,355 extensions are consistently identifiable by our were successfully loaded for each of the five runs. We believe this methods. Considering the direct comparison with Carnus, 180 of Peeking through the window: Fingerprinting Browser Extensions through Page-Visible Execution Traces and Interactions CCS '24, October 14–18, 2024, Salt Lake City, UT, USA

Method Usage Repeated Unique Only 6.1 Ethics and Responsible Disclosure Global APIs 436 432 367 -

- Norm. Stacktrace 436 432 353 0 As we plan to open-source our pipeline to allow for follow-up re-
- Caller & Params 436 423 182 14 search, adversaries could use these fingerprints to identify users.

Variables 359 351 288 79 Thus, in October 2023, we followed best practices in notifications [55, Cookies 25 24 19 14 Storage 85 84 55 43 56] and informed the developers of the Chrome and Firefox exten- IndexedDB - - - - sions reported by our pipeline. We provided the developers with a PostMessages 176 172 138 54 proof-of-concept testbed, allowing them to test and limit the finger- Cross-class 314 305 242 0 printability of their extensions against our vectors in the future [45]. Total 689 682 572 - So far, we sent out notifications for a total of 1,594 Chrome and Table 5: Results for the Firefox dataset 273 Firefox extensions. 30 developers replied to our notification. 16 of them positively acknowledged the underlying issue in their extensions. Six of them did not understand the threat and followed up further. In contrast, four developers mentioned that

security and privacy is not their primary concern or that “it should be the these extensions were not fingerprintable through Carnus’ tech- platform’s responsibility to take care of such issues”. Another four de- niques (i.e., they were not contained in the list shared with us by developers indicated that it is a known problem but also unavoidable the Carnus’ authors). However, their approach relies on a signif- for their functionality (e.g., crypto-wallet extensions). icant fraction of extensions with unique WAR URLs. Therefore, Furthermore, we discussed with three developers who showed Table 4 also shows how many extensions we could detect that interest in understanding the problem in detail as well as finding Carnus could not if randomized runtime identifiers were enabled potential solutions for individual cases. They indicated that they (dubbed No WAR). Here, we find that our approach would still be inject scripts for including third-party libraries (e.g., React libraries), able to fingerprint 484 extensions that Carnus would not be able creating overlays, loading fonts, and so on, which are crucial to to fingerprint anymore. For a fair comparison, we considered only the extensions’ functionalities. They also mentioned the lack of the normalized stacktrace here, as the full stacktrace would also dedicated API (in the current architecture) that injected scripts be affected by randomized runtime identifiers. This highlights that could use to communicate with other extension components (i.e., even if WAR-based fingerprinting becomes infeasible, our attacks content scripts or popups) instead of using the postMessage API. To add significant fingerprinting surface to the state of the art, which summarize, all three developers were unaware that their extensions can also not be overcome by existing defense mechanisms. More were fingerprintable and positively acknowledged our findings. importantly, our fingerprinting techniques cannot be overcome Two of them also affirmed that they will try to reduce the usage of easily by readily-available countermeasures in modern browsers. the fingerprinting vectors we uncovered wherever possible.

5.3 Firefox Dataset 6.2 Limitations Last but not least, we turn our attention to the Firefox dataset. We utilize a hybrid analysis pipeline in this study to detect uniquely Overall, the number of considered extensions is significantly lower identifiable extensions with respect to the newly discovered fin- than the Chrome store datasets of both Raider and Carnus. This is gerprinting strategies discussed in this paper. However, we strictly also observed in the much lower number of extensions that have any note that our tool only reports a lower bound of all the potentially behavior that our attacker model could observe. As with Chrome, identifiable extensions available in the stores due to certain lim- the most potent vectors for Firefox are also stacktraces, which allow itations (and design choices) of our approach. In the first stage, for the detection of 353 extensions. We note that Firefox already we filter out extensions that do not contain a valid declaration of automatically randomizes runtime identifiers. Therefore, Table 5 content scripts, background scripts or WARs with appropriate host also omits the stacktrace row, as we can only rely on the normalized permissions (as discussed in Section 4.2). However, extensions can stacktraces. The variables are the second-most potent vector as 288 also inject or update content scripts from their background, using of these extensions also set global variables in the shared namespace the scripting, tabs, or activeTab permissions. While extensions leading them to be fingerprintable. Our findings highlight that the may request any of these permissions to inject scripts at runtime, potential for fingerprinting through our vectors does not only affect they may only exhibit this behavior on specific hosts. For example, the Chrome extension ecosystem, but the patterns exhibited by the OffShip - Online Shopping Carbon Offsets has script injection capa- extensions also generalize to Firefox. Again, the numbers present a bilities on <all_urls> but only

injects something on the Amazon lower bound as all 572 extensions could be fingerprinted in three and Walmart domains, through the location check at runtime. separate runs (i.e., nine distinct visits). Next, our dynamic step necessitates extensions to exhibit consistent runtime behavior on our test page. This has certain limitations. i.) An extension may cloak its runtime behavior through runtime 6 DISCUSSION logic. ii.) In cases where the extension-injected code executes before In this section, we discuss our disclosure and limitations, followed the test page JavaScript, we do not capture any data from the tests. by an overview of existing defense mechanisms (and why they are This, in particular, is due to the race condition when extensions insufficient for our attacks). Finally, we discuss potential counter- inject code on document_start, and the injected code may execute measures to mitigate the identified vectors. before any script on the attacker page [16]. To not be impacted by CCS '24, October 14–18, 2024, Salt Lake City, UT, USA Shubham Agarwal, Aurore Fass, & Ben Stock

the side effects of race conditions or inconsistent runtime behavior, how to partially stop attacker code from hooking into APIs, discuss we only considered extensions that showed consistent results for how to ensure variables do not leak to the global scope, provide nine visits on our test page and omitted other cases from our find- alternative means of storing client-side information, and finally ings. However, other extensions could still be identifiable through outline how to avoid postMessages being captured. multiple visits to the attacker page, thus, our results present a lower Global APIs. Unfortunately, it would be non-trivial for extension bound of the fingerprintable extensions we uncovered. Also, our developers to prevent observable side effects caused by injected polling approach does not work for extracting IndexedDB data in code, because of the underlying extension architecture. In fact, Firefox since the API to enumerate over all available databases, there are legitimate use cases where extensions may require code i.e., indexeddb.databases, is not implemented for Firefox [37]. injection into the context and the namespace of the visited Web page. Thus, preventing extensions from injecting code will limit their 6.3 Existing Fingerprinting Defenses functionality. To ensure that an attacker-controlled page cannot Prior work has suggested specific mitigation techniques for their hook into APIs called by extension code, the extension developer detected fingerprinting vectors. For instance, creating Shadow or has two options: run all their code which uses the APIs before Parallel DOM, as proposed by Laperdrix et al. [28] and Karami et al. the attacker's code executes or ensure that these APIs cannot be [25], that is inaccessible to the Web page JavaScript, can only pre-overwritten. Note that storing clean references for later use would vent DOM and style-based fingerprinting. Similarly, the preventive most likely again leave traces (as this requires additional variables). strategies by Trickel et al. [59] only tackle DOM-based side effects. To allow for this approach to work, the extension code, therefore, More importantly, none of the existing anti-fingerprinting defense needs to run before the attacker's code. strategies could protect extensions against the set of vectors we pro- An extension can, at the earliest, inject a script into the page posed in this study. This is due to the underlying architecture, such at document_start, i.e., when the browser parses and renders the that the extensions, although in different processes, have shared HTML content. For MV2 extensions, we empirically validated that access to the client-side storage. Moreover, the injected scripts also if an extension injects an inline script (i.e., a programmatically cre- execute in the same JavaScript namespace as the visited Web page. ated script element with an innerText property), this will execute Next, Sjösten et al. [46] suggest randomizing the pointers to the before any page JavaScript. However, the scripts injected through

web-accessible resources included by extensions, which can be embedded in their URL (i.e., the `script.src` property) execute after the page is loaded nowadays through the `use_dynamic_url` key [39]. At the time of our study, we only observe 109 fingerprintable extensions. Fingerprinting of inline scripts is no longer possible since the minimum CSP constraints for content scripts does not allow inline script in extensions to be fingerprinted, even if they all used randomized identifiers [40]. However, extensions can specify that a content script should run in the MAIN world [19] (i.e., is injected directly into removing randomized parts of the URLs within the collected stack-trace of the page). Here, we confirmed that the extension code reliably runs traces that leaves sufficient entropy through line numbers and offsets before the page JavaScript. We can uniquely identify extensions. One can prevent a JavaScript API from being overwritten by freezing their native definition through the `Object.freeze` API, 6.4 Extension Ecosystem & Standards thus, maintaining the integrity of respective APIs [41]. Extension developers could use this mechanism to freeze the native definition of Web Store have certain restrictions and built-in protection mechanisms in place to protect against many critical security vulnerabilities in the MAIN world, before any page JavaScript executes. This would prevent the attacker from capturing any execution traces at runtime. The features introduced, such as blocked remote-code inclusion and strict CSP rules, may help limit security issues on the client with global JavaScript objects (i.e., `Array.prototype`, `String.prototype`, side that originate from extensions. However, extensions can still etc.), while the window APIs and properties (i.e., `postMessage`, `setTimeout`, etc.) cannot be frozen. We extracted the unique features often inject code that executes in the applications' context, thus that we collected for 1,769 extensions fingerprintable through the causing observable side-effects. This supports our findings, given global APIs from the Raider dataset. We found that 829 of these will that 1,611 / 2,747 of the fingerprintable extensions we detected are, still be fingerprintable after freezing all possible JavaScript APIs in fact, Manifest V3 standards. Besides, our proposed fingerprinting in the global namespace. We note that freezing global JavaScript strategies are not limited to the Chrome extension ecosystem and objects might also cause unintended side-effects to benign websites also apply to Mozilla Add-ons and other extension stores. We show which may extend or overwrite these APIs for their functionality. the versatility of our approach by running our tests on Firefox extensions. Further, developers could avoid fingerprinting through global variables by either scoping them appropriately (e.g., through architecture is similar across the extension ecosystems [35]. the `var` keyword) or wrap the injected code within an Immediately Invoked Function Expression (IIFE), since the execution context of 6.5 Recommendations and Mitigation Strategies an IIFE is destroyed right after it executes. This way, no function In this section, we discuss ways in which developers can avoid exposing fingerprintable behavior to an attacker. We first discuss 3 More details on the tests at <https://raider-ext.github.io/raider/tests/> Peeking through the window: Fingerprinting Browser Extensions through Page-Visible Execution Traces and Interactions CCS '24, October 14–18, 2024, Salt Lake City, UT, USA

definitions would pollute the global scope. Naturally, if an extension components, including `postMessages`. They built honeypages, based needs to register global variables for interaction with a page, developers on the extensions' description provided by the developers, to trigger operators might intentionally expose variables. Therefore, any change gets the extensions' runtime behavior. In 2022, Solomos et al. [49] in such exposure would imply a negative effect on the functionality. leveraged the `MutationObserver` to capture any DOM modifications attributed to individual extensions during execution. In fact, previous Storage APIs. While extensions may need to store runtime information only compared the DOM before/after execution, thus information related to visited websites (e.g., preferred UI settings), this missing the invisible and transient interactions that happen during can serve as a source of entropy for tracking users. In this case, extensions but are not observable after execution anymore. Finally, extension developers should utilize the `chrome.storage` API instead they also showed that user-induced runtime events, such as keyboard (by replacing the `localStorage` invocation with `chrome.storage` board and mouse events, could increase the extensions' interaction in the code), where the storage container is accessible only to individual extensions. Further, suppose an extension needs to store a large chunk of data on the client side, i.e., in the `IndexedDB`. In that case, we focus on a set of fingerprinting vectors that are agnostic to DOM-based side effects. We show that the execution of extensions' injected scripts in the realm of the Web context [9, 12]. Similarly, we recommend that developers of Web application could leave traces in the global JavaScript namespaces do not set cookies on arbitrary domains since an attacker can also pace (e.g., global variables). This is because the attacker JavaScript observe them through JavaScript or in an incoming request to the can overwrite the global JavaScript APIs and properties to capture attacker-controlled server. their invocations, which then adds to the fingerprinting surface for `Messages`. Assuming the extension can inject its code before the extensions [29]. Moreover, we highlight that any interactions with page JavaScript executes, it can ensure that `postMessages` are only the client-side Storage APIs and other global JavaScript APIs in the relayed after filtering out the extension-sent ones. Specifically, applications' context further aid in fingerprinting extensions. the extension can overwrite the `addEventListener` API and win- In 2017, Sjösten et al. [47] showed that browser extensions are `chrome.web_accessible_resources` property to ensure that if an event handler is registered through the web-accessible resources (WARs) they interested, the handler is only invoked if a message does not originate from different websites. In 2019, they further found that tracking from the extension. Doing so, the attacker will be unable to recover ing websites could probe for these included resources even when extension-sent messages. However, this leads to an overwritten the browser randomizes the extension runtime identifier used to global API, which could be used as a vector for fingerprinting yet fetch these resources [46]. Fortunately, the current extension architecture mitigates any attempt of WAR-based fingerprinting by the same approach to ensure a greater anonymity set. opting into dynamic URLs [39]. Around the same time, Starov et al. [52] showed that unnecessary code bloats within extensions could Separated execution of extension-injected scripts. Orthogonally, similar to `ShadowDOM`, extensions could also have access to a "Shadow showed that extensions could also be fingerprinted based on the `chrome.Window`" object, which

would provide them with a separate stylesheet injection patterns observable from Web pages. Overall, global namespace. Thus, any extension-incurred changes (e.g., vari- these fingerprinting techniques are orthogonal to the set of vectors able registration, API usage, etc.) would not be visible to the Web we introduce in this study. In fact, we investigate the observable page JavaScript. This aligns with the ongoing discussion among side-effects of the execution of extension resources, even when the stakeholders of the WebExtensions framework [60]. Overall, the attacker application cannot probe for an extension's existence we also urge browser vendors to take the necessary steps toward because of security measures in place at runtime. upgrading the isolation boundaries of the extensions in this regard. Unfortunately, evaluating the unintended side-effects of this stricter Browser Extension Fingerprinting Defenses. In 2019, Trickel et al. isolation approach would be non-trivial, as it is extremely challeng- [59] proposed a mitigation technique to counter DOM-based exten- ing to automatically infer whether an extension developer actually sion fingerprinting. By randomizing the DOM element identifiers, wanted to interact with the page's global object or not. an attacker can no longer attribute them to individual extensions. In 2022, Karami et al. [25] suggested having a separate copy of the 7 RELATED WORK actual DOM, a Parallel DOM, for page-based interactions vs. a User Browser Extension Fingerprinting. In 2017, Starov and Nikiforakis DOM for the extension-based interactions and inaccessible to the [53] first quantified the fingerprinting characteristics of Chrome ex- Web page. This is similar to the concept of Shadow DOM, proposed tensions based on the extensions' interaction with the DOM. They by Laperdrix et al. [28] in 2021, to isolate the website's view and the instrumented the content scripts of the extensions to create the re- extensions' view of the DOM. In 2022, Solomos et al. [48] suggested quired DOM structure on the fly, and they captured the extensions' using the isTrusted property of events when listening to them to runtime interactions with the DOM. However, this fingerprinting avoid side effects of fake events dispatched by the attacker. Unfortu- strategy does not work for content scripts now, as they share a nately, none of these defense strategies protect extensions against different DOM handle and are not accessible to the Web page [34]. the set of fingerprinting vectors we discuss in this paper. This is be- In 2020, Karami et al. [24] automated Chrome extensions' finger- cause extensions may cause observable side-effects on the window printing based on their interaction with the DOM and through (e.g., window.localStorage or globally-accessible variables), even their communication patterns with different client- and server-side observable to an attacker with a restricted view of the DOM. CCS '24, October 14–18, 2024, Salt Lake City, UT, USA Shubham Agarwal, Aurore Fass, & Ben Stock

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Roberto Developer Tools 600 Just for Fun 52 Perdisci, and Wenke Lee. 2015. Understanding Malvertising Through Ad-Injecting Tools 270 Privacy & Security 39 Browser Extensions. In WWW. Accessibility 174 Education 27 [62] Jianjia Yu, Song Li, Junmin Zhu, and Yinzhi Cao. 2023. CoCo: Efficient Browser Shopping 144 Communication 22 Extension Vulnerability Detection via Coverage-guided, Concurrent Abstract Social Networking 132 Functionality & UI 14 Interpretation. In CCS. Productivity 72 Social & Communication 12 Art & Design 8 Entertainment 7 News & Weather 7 Well-being 2 A APPENDIX Photos 2 Games 2 Household 1 Travel 1 1 // API Type 1 Table 6: Categories of extensions reported by Raider to be 2 var foo = [1, 2, 3]; fingerprintable. We could not extract any explicit category 3 foo.forEach((element) => { 4 console.log(element); for 30 extensions from the Chrome Web Store. 5 }) 6 // API Type 2 7 var bar = new Set([1, 2, 3]); 8 console.log(Array.isArray(bar));

Listing 5: JavaScript APIs and their execution contexts.